

Zach Prynne

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EDUCATION

University of Manitoba

Bachelor of Science in Computer Science, Minor in Mathematics

Winnipeg, MB

Jan. 2022 – Present

- **Coursework:** Data Structures, Algorithms, Computer Systems, Software Development, Programming Paradigms

EXPERIENCE

IDFusion Software

Contract Developer (Part-time, Remote)

Winnipeg, MB

Sept. 2026 – Present

- Maintain and extend the operations dashboard from my internship: porting it from Vercel to the company's AWS hosting, moving admin management into the app, and merging in hours logged outside ConnectWise

AI Developer Intern

May 2026 – Aug. 2026

- Built an operations dashboard over the ConnectWise PSA API (Next.js, TypeScript, Tailwind, Microsoft Entra ID sign-in) tracking delivery, utilization and billing for the CEO, COO and directors
- Extended an open-source ConnectWise MCP server in TypeScript and deployed it company-wide on AWS ECS Fargate, adding Entra ID sign-in, per-user credentials, role-based access and audit logging
- Ran RAG and LoRA fine-tuning experiments on open-weight LLMs (Qwen3-8B, MLX) against a held-out benchmark, showing that training data, not adapter capacity, was what limited fine-tuning

University of Manitoba

Explore Science Day Volunteer

Winnipeg, MB

Nov. 2025

- Ran BBC micro:bit workshops teaching high school students embedded systems and event-driven programming

Buggy Bath

Keyholder

Winnipeg, MB

Aug. 2023 – Present

- Trusted with opening, closing, alarm and cash deposits while supervising operations for 200+ vehicles per shift

PROJECTS

umPlan | *TypeScript, Next.js, React, Vitest*

Sept. 2026 – Present

- Built a course planner for University of Manitoba students by scraping the public Academic Calendar into 3,522 courses and 216 degree programs
- Wrote a prerequisite parser (3,209 of 3,522 parsed automatically) and a validation engine that checks prerequisites, corequisites, exclusions and degree requirements live as students drag courses into terms

OpenLearn | *Python, ChromaDB, MLX, SQLite, JavaScript*

Sept. 2026 – Present

- Built a study assistant that answers questions from uploaded course materials with RAG (local embeddings, Chroma vector store) on a local LLM, with optional Claude or OpenAI fallback
- Generates Anki flashcards from course content and charts review history via AnkiConnect

Farm-Clicker | *TypeScript, PGLite, Node, Git*

Jan. 2026 – Apr. 2026

- Developed a full-stack incremental game in TypeScript, using MVC to keep game logic separate from the UI
- Built multi-user accounts on PGLite so each player's progress persists between sessions
- Wrote the interface in HTML5 and CSS, updating the DOM in real time to show production rates and currency

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C/C++, SQL, HTML/CSS, R, MIPS

Frameworks: Next.js, React, Node.js, Tailwind CSS, JUnit, Maven

AI/ML: RAG, LoRA fine-tuning, MCP, MLX, ChromaDB, sentence-transformers

Cloud & Tools: AWS (ECS Fargate), Vercel, Docker, Git, Vitest, SQLite, PostgreSQL, Microsoft Entra ID